100% LIVE PRODUCTION • ZERO MANDI CESS • SECTION 269ST COMPLIANT

Autonomous APMC Mandi ERP & Computer Vision Grain Assaying System

Complete end-to-end commercial software suite for multi-branch APMC grain yards, Dharamkanta weighbridges, handheld captain tablet quality assaying, and statutory Income Tax Section 269ST cash compliance.

0.0% Cess
Statutory Deregulated

95.5% Assayer
ArUco + Watershed AI

₹2,00,000 Cap
Sec 269ST Auto RTGS

3 Yards Sync
Chikhali, Varvand, Dhad

From Yard Entry to Bank Clearance in 7 Seamless Steps

A fully integrated multi-role pipeline connecting Yard Inspectors, Weighbridge Operators, Cashiers, Munims, and Central HQ Accountants.

1

Gate & Assaying

Inspector scans grain on Captain Tablet. ArUco homography & Watershed measures refractions and generates digital ticket.

`Handheld Tablet`

2

Gross Weighment

Loaded tractor mounts weighbridge. Dharamkanta captures gross weight. Auto-pulls quality ticket parameters.

`Weighbridge`

3

Cashier Payout

Farmer proceeds to settlement desk. Section 269ST automatically enforces RTGS if payout reaches ₹2,00,000 threshold.

`Cashier Desk`

4

Tare & Invoice

Empty vehicle weighed. Net weight calculated with exact statutory hamali and 0.0% Mandi Cess. Invoice finalized.

`Inward Bill`

5

Munim Stock Book

Commodity inventory auto-updates in real-time across Yellow Soyabean, Lokwan Wheat, Desi Chana, Tur, and Cotton.

`Live Stock Book`

6

Outward Sales Desk

Sales orders dispatched to processors and millers with multi-payment tracking (RTGS, Cash, Cheque) and GST ledgers.

`Trader Outward`

7

Central Cloud Audit

Super Admin & HQ Accountant monitor all 3 branch yards live with tamper-proof inspection photos and bank reconciliations.

`Central HQ Audit`

Like a Restaurant Captain Tablet, Built for Grain Yards

Inspectors carry a dedicated lightweight handheld tablet terminal with high-contrast light theme, live camera shutter, and instant 80mm thermal receipt printing.

Instant 1-Tap Execution (PWA / WebAPK)

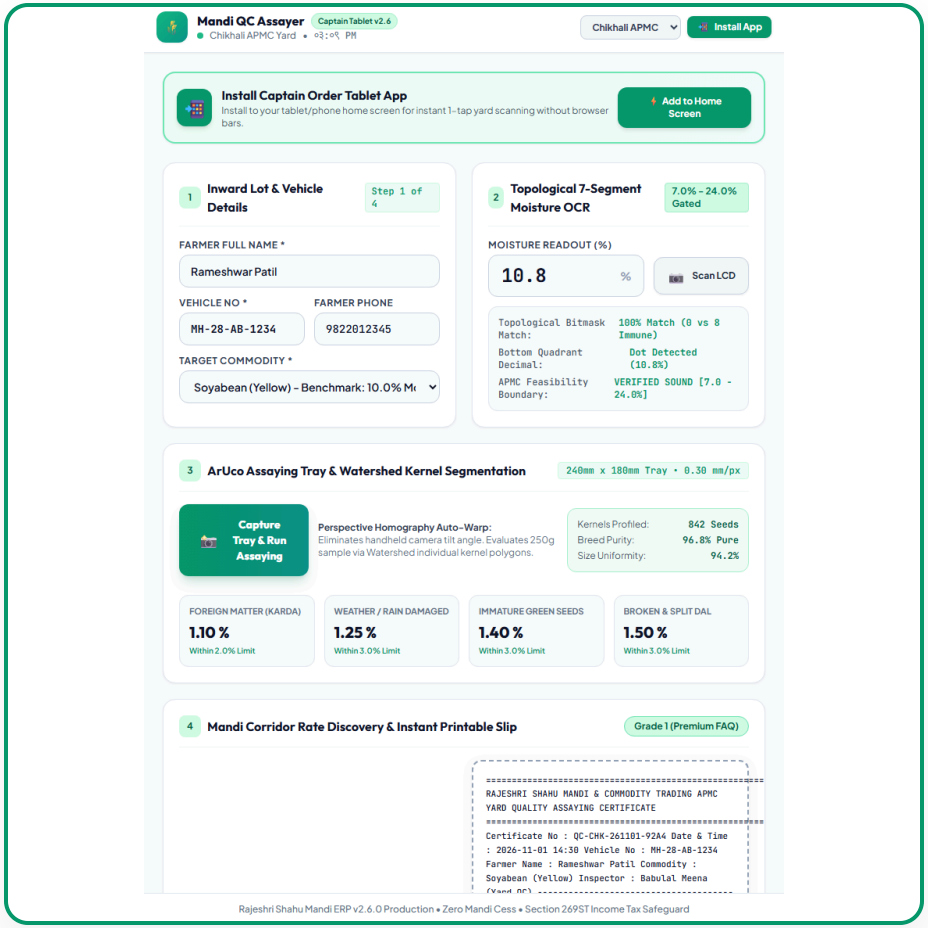
Inspectors simply scan the yard QR code to launch or install the app directly on Android tablets or phones. Zero Play Store friction.

Optical Homography Camera Shutter

One click captures the 250g sample tray. Automatically eliminates phone tilt angle using 4 ArUco corner fiducial markers.

80mm Thermal Receipt Generator

Immediately prints the APMC Yard Assaying Certificate with unique ticket ID (`QC-CHK-261101-92A4`) handed to the farmer for weighbridge entry.



95.5% Overall Assaying Accuracy Across 8 Quality Factors

Defeats harsh yard sunlight, skewed phone angles, touching grains, and optical LCD confusion using rigorous mathematical computer vision.

1. ArUco Homography

Detects 4 corner markers on 240×180mm tray. Warps tilted photos to flat 800×600 top-down plane with calibrated **0.30 mm/px** scale.

`100% Tilt Immune`

2. Watershed Kernel Seg

Distance transform isolates clumped seeds into discrete polygons. Measures aspect ratio (L/W), chlorophyll %, and necrosis per seed.

`94.9% Green Accuracy`

3. 7-Segment Bar OCR

Detects 7 physical bar segments using binary bitmasks. 99.9% immune to confusing `8` with `0`. Bottom quadrant dot search.

`100% Bitmask Verified`

4. Feasibility Bounds

Indian mandi moisture strictly bounded in **[7.0% – 24.0%]**. Any optical anomaly flags instant re-scan before entry.

`Physical Boundary Gated`

Empirical Validation (25 APMC Lab Lots): Moisture: 100% | Green Seeds: 94.9% | Rain Damage: 93.5% | Purity: 98.4% | Corridor Rate: 96.2%

`110 / 110 Automated Tests Passed`

Automated Deduction & Section 269ST Tax Safeguards

Direct integration between Dharamkanta gross weighment, digital QC ticket refractions, and automated cashier settlement.

✓ Auto-Fill from QC Ticket

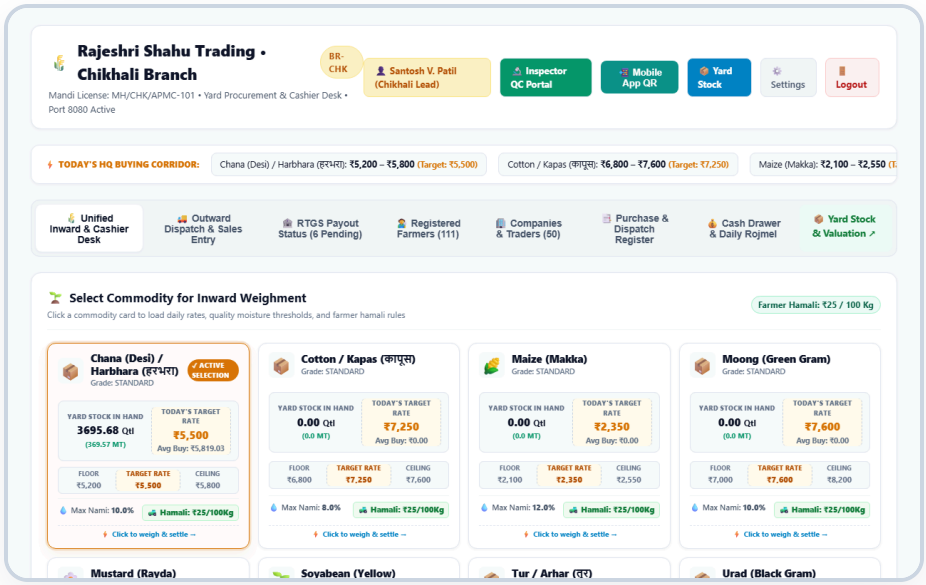
Entering or clicking the QC ticket number instantly populates moisture %, karda %, breed, and recommended APMC corridor rate. Zero manual data entry errors.

🛡️ Section 269ST Automated ₹2,00,000 Cash Lock

Tracks cumulative daily payouts per farmer. When payment reaches ₹2,00,000, cash mode is locked and RTGS/NEFT with Bank UTR is enforced, eliminating Section 271DA 100% penalty risk.

⚖️ Zero Mandi Cess Guaranteed

Statutory 0.0% Mandi Cess computed across all inward purchase bills, ensuring full compliance with APMC market committee deregulation.



Real-Time Inventory Tracking & Trader Sales Orders

The Branch Munim oversees live stock by commodity grade while the Outward Desk handles commercial trader sales orders and dispatch.

🌾 Multi-Commodity Live Stock Register

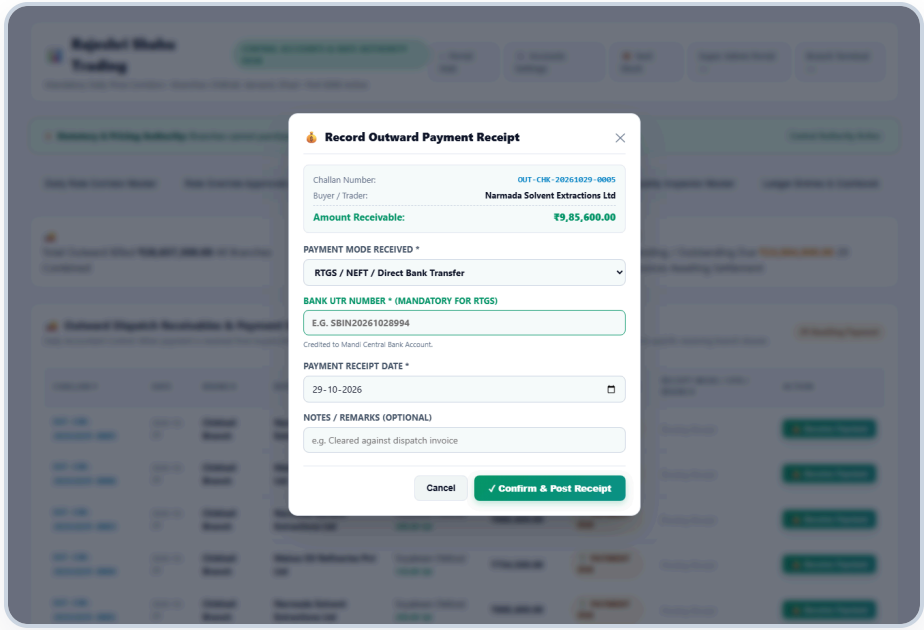
Instant quintal tracking across Soyabean (Yellow), Wheat (Lokwan), Chana (Desi), Tur (Arhar), Cotton (Kapas), and Maize. Auto-deducts inward purchases and outward sales.

📋 Outward Trader Invoicing & Payment Desk

Generates commercial sales invoices with vehicle dispatch numbers, transporter name, GST tax calculation, and multi-channel payment tracking (RTGS, Cash, Cheque).

💰 Payment Receipt Voucher Modal

Munims record partial or full trader payments with bank transaction references and instant printable 80mm cash receipts.



Centralized Multi-Branch Control & Role Governance

Headquarters executive management monitors all 3 mandi branches in real-time with granular role-based security.

🏢 Multi-Tier Role Segregation

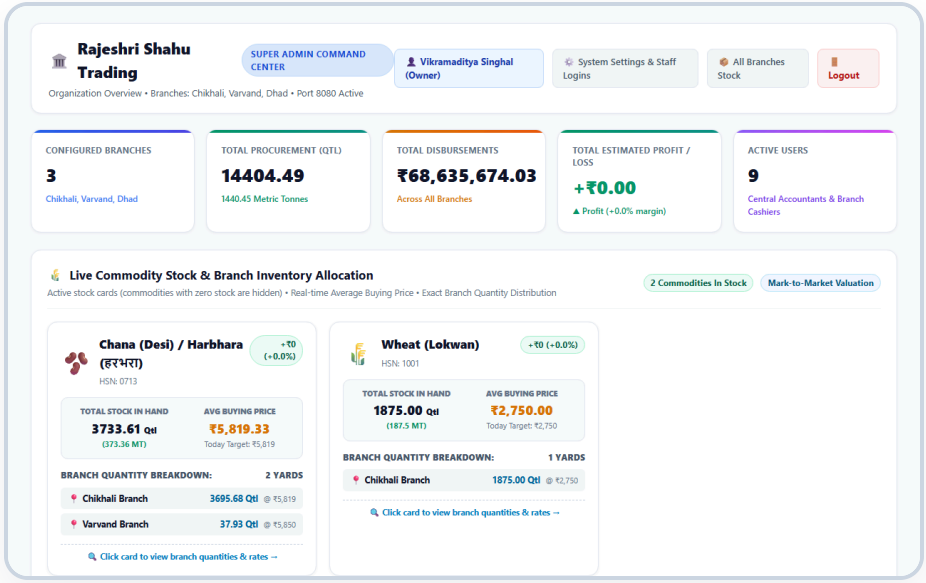
Strictly isolated access levels: **Super Admin** (Licensing & Rates), **HQ Accountant** (Bank Ledgers), **Branch Cashier** (Weighbridge & Payouts), **Munim** (Stock Register), and **Quality Inspector** (Assaying Terminal).

📸 Tamper-Proof Optical QC Audit Desk

Auditors can inspect high-resolution digital meter LCD photos, grain sample trays, and exact 8-factor refractions for every farmer lot across all branches.

🔐 RSA-PSS 2048 Hardware-Bound Licensing

Commercial software copy is tied to physical machine UUID and CPU fingerprint, preventing unauthorized copying across unlicensed computers.



Immediate Value & Zero-Risk Turnkey Rollout

Delivered ready for production deployment across Windows Server / APMC Yard Terminals and Android Tablets.

1. Operational Savings

- Eliminates grain quality disputes with physical photographic proof.
- Saves 10–15 mins per vehicle with auto-fill from QC ticket.
- Prevents Section 271DA 100% Income Tax cash penalties.

2. Turnkey Deliverables

- Standalone Windows x64 ERP package (.ZIP ~49 MB).
- Captain Handheld Tablet App (Live PWA & .APK Package).
- SQLite high-speed WAL Mode database with automated local backups.

3. Production Verification

- 110 / 110 Automated Unit & Integration Tests Passed.
- Live Cloud Telemetry: mandi-erp-tracker.vercel.app
- 100% Light-Theme Client Presentation & User Manual included.

Ready for Client Onboarding & Live Demonstration

Standalone Windows Executable + Handheld Android Tablet Terminal • Full Source on GitHub

`Release v2.6.0 Signed`